

Identifying the Mechanism Through Which Private Prisons Increase Sentencing

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1. Abstract

This paper attempts to identify the mechanism through which private prisons cause increased judicial sentencing, an effect which has been confirmed to exist by previous research. Of primary consideration is how the existence of minimum occupancy clauses (guaranteed payment for a certain level of prison occupancy) changes a private prison's effect on sentencing. The incorporation of minimum occupancy clauses has direct implications on both the cost saving hypothesis and the judicial capture hypothesis, the two most likely candidates for the cause of the private prison effect. Using a combination of theoretical models and empirical data analysis, I find that the effect of minimum occupancy clauses is exactly in line with what the theory behind the cost saving hypothesis would predict, implying that it is the stronger candidate for why private prisons lead to increased sentencing.

2. Introduction

2.1 Background and Previous Work

In his sixth day in office, President Joe Biden signed an executive order to phase out the use of private prisons in the Department of Justice, citing the harm of profit-based incentives to incarcerate. Between 1980 and 1994, the private prison population increased nearly 200%, in part due to the war on drugs (Pfaff, 2015). The United States spends roughly \$4 billion out of a total \$84 billion a year on private prisons to incarcerate 8.4% of the 2.3 million prisoners (Bureau of Justice Statistics).

Private prisons market themselves as a cost saving alternative for States and are appealing to address budget issues and overcrowding. However, these cost-saving claims may be illusory as some states, like Arizona, have found private prisons to be more costly than state prisons (In the Public Interest, 2013). In addition to the cost issues, previous research has highlighted other concerns, such as private prisons shirking on quality of rehabilitation which increases the rate of recidivism, and their negative effect on the implementation of the judicial system.

The concern about decreased quality has been analyzed theoretically and empirically. The incentive to minimize costs paired with minimum occupancy clauses that guarantee revenue alleviate the need for private prisons to compete on non-contractible quality, enabling them to shirk (Hart et al., 1997). Empirical analysis has found that private prisons do not have the incentive to retain experienced custody staff (Camp and Gates, 2006). In addition, private prison inmates in Florida and Minnesota, were found to have a statistically

significantly higher risk of recidivism with evidence pointing towards fewer visitation and rehabilitation programming opportunities for the incarcerated offenders in private facilities (Spivak and Sharp, 2008; Duwe and Clark, 2013).

When investigating the impact on the judicial system, recent empirical analysis found that private prison openings have a significant effect on sentencing in the judiciary where the prison was opened, an impact not seen with public prison openings (Dippel and Poyker, 2020; Galinato and Rohla 2020). Two potential explanations for this effect are that private prisons directly lobby for increased incarceration levels, and that judges internalize the cost saving benefits associated with private prisons.

2.2 Cost Saving vs Judicial Capture

As the ones ultimately handing down sentences, the incentives of the judges are of utmost relevance. Any change in sentencing caused by a private prison must flow through the judges in some way. As indicated by Dippel and Poyker (2020) there are two potential motives: the judicial capture motive and the internalization of cost savings. Judicial capture entails prisons directly or indirectly lobbying judges for increased sentencing, which will lead to higher revenue for the prison. This is what occurred in the 2008 Kids for Cash scandal, where two judges in Pennsylvania accepted bribes in exchange for levying harsher sentences on juveniles being detained at a private detention center.

Cost savings internalization is when judges' decisions are influenced by the perceived marginal cost of incarceration. When an additional bed incurs a large additional cost, internalization would lead to decreased sentencing. Private prisons market themselves as providing incarceration services at a lower cost than a public prison could. Judges may then internalize the notion that private prisons are associated with a lower marginal cost of sentencing and increase sentencing accordingly when a new private prison opens.

2.3 Paper Objectives

This paper seeks to shed light on which of the two potential mechanisms for private prison's effect on sentencing is the more likely. An additional component of private prison operations that has not yet been considered in the literature is the effect of minimum occupancy quotas. Many private prison contracts include a clause assuring that the government will pay for a certain percentage of their prison beds, regardless of whether these beds are actually filled. As shall be seen, under both the cost saving hypothesis and the judicial capture hypothesis, the addition of a minimum occupancy clause has direct implications on the mechanism

through which private prisons would incentivize increased sentencing. Analyzing how the effect of private prisons on sentencing differs between those with minimum occupancy clauses and those without is an effective way to gain insight as to what is the driving force behind the private prison effect on sentencing.

3. Theoretical Models

3.1 Cost Saving Hypothesis

To test the cost saving hypothesis, the theoretical model is built upon the sentencing enforcement model derived by Galinato and Rohla (2020). In this model, the sentencing authority determines an optimal sentence $S(g)$ by maximizing societal welfare W , where:

$$W = U(S, g) - d(S) - pS$$

S is the final sentence, p is the price of incarceration, $U(S, g)$ is the societal utility gained from sentencing an individual who committed a crime of severity g to a sentence S , and $d(S)$ is the societal disutility associated with incarcerating this individual. Included in the disutility function may be many factors such as the decrease in the available labor force, the long-term effects on children left without a parental figure, financial effects on a spouse who may have relied on them as a primary breadwinner, and other factors. The first order condition is:

$$U_S(S^*, g) = d_S(S^*) + p$$

where S^* is the optimal sentence for a criminal who committed a crime of severity g . If for a given level of crime severity g , $U_S(S, g) \leq d_S(S) + p$ for any level of sentencing S , then probation is more optimal than incarceration. The incarceration rate is then defined as:

$$I(g^I) = \int_{g^I}^{\infty} S(g) dg$$

where g^I is an endogenously determined point of indifference to the sentencing authority between incarceration and probation and is the point where $U_S(0, g) \leq d_S(0) + p$ holds with equality.

The rate at which higher sentencing increases utility is decreasing ($U_S > 0$ and $U_{SS} < 0$) i.e. a decreasing marginal utility of sentencing. The optimal level of sentencing is determined by $U_S(S^*, g) = d_S(S^*) + p$ meaning that a lower level of p allows for a lower marginal utility of sentencing S at optimum, and thus a higher S^* , since the marginal utility of S is decreasing in S .

Under the cost saving hypothesis, it is believed that $p_{private} < p_{public}$, i.e., private prisons provide the same incarceration services as public prisons, at a lower cost. Under this hypothesis, the decrease in p advertised by private relative to public prisons is the driving force behind the increase in sentencing caused by private prisons.

A special scenario occurs within the framework of this model when minimum occupancy clauses are considered. Where minimum occupancy clauses are in place, the marginal price p is effectively set to 0 exogenously.¹ With $p = 0$ the optimal level of sentencing S^* is thus the point where $U_S(S^*, g) = d_S(S^*)$ which allows S^* to rise to a new equilibrium where the marginal utility of sentencing is exactly equal to the marginal disutility of sentencing. Although the same overall amount of money for these beds is being paid to the private prison, the price point of each bed is removed from the equation that is maximized by the sentencing authority, at least until the quota is reached. With $p_{clause} = 0 < p_{private} < p_{public}$, the model suggests that private prisons with minimum occupancy clauses cause an even larger increase in sentencing than a private prison without one.

3.2 Judicial Capture Hypothesis

Another model is needed to explore the situation where judicial capture, or corruption, is the cause of increased sentencing. In this scenario, it is assumed that the judge does not weigh the explicit financial cost of sentencing in his decision-making process² and that the increase in sentencing seen from private prisons is due to other factors, namely corruption and bribes.

Altering somewhat the model developed by Grossman and Helpman (1994), the enforcement authority determines an optimal sentence S^* according to the welfare equation:

$$W = U(S, g) - d(S) + aBS$$

Here $U(S, g)$ is again the societal utility from incarcerating a criminal who committed a crime of severity g to sentence S , and $d(S)$ is the societal disutility associated with incarcerating an individual for sentence S .

As a brief aside, it is logical to conclude that if a judge were not to weigh the financial cost of incarceration in his decision, he very well may not weigh the social disutility into his decision either, especially since the

¹ The judge who determines S is certainly not involved with the decision-making process of whether to implement a minimum occupancy clause but is the one who chooses S to maximize the welfare function.

² A corrupt judge wouldn't care!

money to pay for price p comes from public funds, i.e., tax dollars, so in a sense the price is simply another component of public disutility. With that thought process in mind, in this model one can instead think of $d(S)$ as a risk function associated with passing a sentence above what would be considered fair. There are precedents, as well as explicit laws and regulations, guiding what sentence is ultimately passed by a judge. One can imagine a “fair” level of sentencing that a neutral party would assume would be passed for a specific crime. The further above this “fair” level a judge’s sentencing decision rises, the more likely observers will become suspicious, a judge’s reputation will be tarnished, and he may even be exposed and thus lose his source of income. To avoid this, a judge maximizes the benefit received from a higher bribe while considering the increased likelihood that he will be discovered associated with a higher sentence. Both specifications, disutility and risk of exposure, produce the same results, and the intuition is largely similar as well.

The additional component to this model not seen previously is aBS , where B is the bribe offered to the judge by the private prison firm for increased sentencing, and a is the judge’s level of corruption, or willingness to take a bribe³. The first order of condition is then:

$$U_S(S^*, g) = d_S(S^*) - aB$$

Recall that in the previous model, a positive p raised the marginal utility point needed to reach optimality, and thus decreased the optimal level of S^* since the marginal utility is decreasing. The reduction or complete removal of p is what caused the increased sentencing effect. The effect of $-aB$ seen in this model is thus opposite. A negative aB lowers the marginal utility pointed needed to reach optimality, and thus increases the optimal level of S^* , since the marginal utility is decreasing. Under this model, bribes paid by private prison lobbyists lead to increased levels of sentencing. It is clear that when $a = 0$ (a perfectly uncorrupt judge) there is no change from above. A judge with a level of corruption $a = 0$ will choose a level of S^* where

$$U_S(S^*, g) = d_S(S^*)$$

To incorporate the effects of the minimum occupancy clauses into this model, some additions must be made. I first specify a utility function maximized by the private prison corporations, or more specifically, their lobbying or advocacy division. For the purpose of this analysis, I assume that only one group within the private prison is responsible for increasing revenue through lobbying and advocacy, and thus the same group that

³ It is certainly possible that the bribe structure would not operate in this simple linear form, with the private prison lobbyists offering a standard price per increased year of sentencing. It may be that lobbyists offer a specific price for a specific desired sentence, and the judge can choose to take it or leave it. That said, a longer sentence would certainly require a higher price and any cost function $B(S)$ that is increasing in S will produce essentially the same results, so for the sake of mathematical clarity this specification will suffice.

would have to exert time and resources to push for minimum occupancy clauses would also be the ones that decide on optimal bribes to be paid⁴. Their utility function is thus specified as:

$$U(c, B) = p * S_{no}(a, B) + m(p) * L_i(c) - BS_{no} - c$$

$$\text{Subject to: } H_i = c + B$$

Here, p is the price paid to the company per prison bed with no minimum requirement, or for beds above the minimum requirement. S_{no} is the sentencing beyond what is included in the minimum occupancy clause (equal to S when there is no minimum occupancy clause in place), and is determined by the judges level of corruption a and the bribe paid B . $m(p)$ is the lump sum payment associated with the enactment of a minimum occupancy clause at an institution with bed price p ⁵, $L_i(c)$ is the likelihood function that a minimum occupancy clause will be enacted given lobbying expenditure c with respect to private prison i ⁶. H_i is the total budget available for lobbying a state government with respect to private prison institution i .

This framework illuminates a few interesting concepts. The first is clearly that different prison institutions will have different optimal levels of B and c , depending on p , a , and $L_i(c)$ for that institution and judge. Certain states such as Ohio, Colorado, and Arizona, which are known for having high levels of minimum occupancy quotas, most likely have a high $L_i(c)$, meaning that a minimum occupancy clause can be enacted with relatively less lobbying expenditure in these states. It is, however, also possible that these states have judges with a low a (a low responsiveness to bribe B) and it is thus relatively more valuable to the lobbying group to expend a majority of their budget H_i on passing a minimum occupancy quota as opposed to lobbying judges directly. Vice versa, states that may be lauded for resisting the installment of minimum occupancy quotas are assumed to have a low $L_i(c)$ but may instead actually have quite corrupt judges with a high level of a , so it is more cost effective for the prison lobbyists to increase sentencing through direct bribes instead of pushing for quotas.

The second and more important thing to note is that under this framework, it is possible for minimum occupancy clause private prisons to increase sentencing by less than it would be without such a clause, while still being the optimal decision for the private prison firm. Assume that, as the model states, lockup quotas

⁴ Note that bribes need not be as explicit as paying a judge directly for passing a harsher sentence but could be accomplished through donations when a judge is up for re-election or other implicit forms of corruption.

⁵ This may not always be equal to p times the number of beds covered by the contract, as some discounts or other structures may be factored in, but is increasing in p .

⁶ i could also be thought of more generally as a state, or a state-prison pair. Different state governments will have different attitudes towards private prisons, and even within a state different prison openings and contract negotiations will have different scenarios surrounding them, so this function will not be the same for every institution within a given state.

are enacted at institutions where the judges have relatively low a and the state government has relatively high $L_i(c)$, and judges are bribed explicitly at institutions where judges have relatively high a and state governments have relatively low $L_i(c)$. At the minimum occupancy clause institutions/states, the judges responsiveness to bribes will not change once the occupancy quota is enacted, and it will still not be cost efficient to bribe these judges for sentencing above the minimum level or beyond capacity. At these institutions, sentencing will initially increase with the enactment of the quota, but beyond that there is no added incentive. Sentencing will stay at normal rates for the remaining prison capacity.

At institutions where explicit bribes are the most efficient, minimum occupancy clauses are hard to enact and judges are more corrupt. Because of this there is no hard cap where the judge's incentive to increase sentencing disappears, at least until the prison budget is exhausted with $B = H_i$. Due to this, sentencing may be enhanced (according to a judge's responsiveness to bribes) all the way to 100% capacity and even beyond⁷. With responsive enough judges, bribes could be made to overflow prisons far past their intended capacity.

The budget H_i can also be different in different counties or states. If a particular group of judges is identified as being exceptionally susceptible to bribes, budget could be reallocated towards these states to increase H_i . Leftover unspent funds in states where minimum occupancy clauses were enacted (high $L_i(c)$), but individual judges remain stalwart (low a) would likely be the first funds to be moved to areas where a was high, as there is no more use for them in low a states once the minimum occupancy clause has been successfully enacted. This reallocation effect would then lead to $H_{low a} < H_{high a}$ meaning more funding in states with corrupt judges, exacerbating the effect. In this case there would be a stronger increase in sentencing because of private prisons with no minimum occupancy clauses as opposed to those with clauses.

3.3 Discussion

Two models that incorporate the effect of minimum occupancy clauses into sentencing behavior have now been examined. Under the cost saving hypothesis, lower costs lead to increased sentencing, so $p = 0$ because of a minimum occupancy clause would cause the greatest increase in sentencing. Under the judicial capture model, the effect is potentially more ambiguous, but likely creates the opposite effect. While minimum occupancy clauses are still an effective way for private prison firms to maximize revenue in certain states, due to the reasons described above their existence may not incentivize higher sentencing as much as the mechanisms in play where they do not exist. Thus, if a stronger effect on sentencing due to private prisons

⁷ Many private prisons routinely run at 150%+ capacity.

with minimum occupancy clauses is observed, this is evidence in favor of the cost saving hypothesis as the driver behind private prison's stronger effect on sentencing than public ones. If a weaker effect of minimum occupancy clause prisons is observed, this is likely evidence towards the judicial capture motive as the driving mechanism behind the private prison effect.

4. Data

To study this effect empirically, our goal is to compare the length of a given sentence (the intensive margin) and the total number of people sentenced in a given state and year (the extensive margin) with the capacity of public prisons, private prisons without minimum occupancy clauses, and private prisons with minimum occupancy clauses in a given state and year. Two datasets were created. The first contains federal sentencing data obtained from the Monitoring of Federal Criminal Sentences data series maintained by the United States Sentencing Commission and hosted by the ICPSR. This annually published dataset contains a record of every federal criminal sentence each year. It includes information such as the state in which the trial took place, severity of the crime, the race, sex, and age of the defendant, and the total number of months he or she was sentenced to. Data from 1999 through 2015 was included in the final dataset that was used. The second contains state level sentencing outcomes from several states and was aggregated by utilizing individual states public records requests processes. Different states have different regulations concerning the data that is available, whom it is available to, and the timeline of obtaining the data. State sentencing data was ultimately collected from Arkansas, Georgia, Maryland, Texas, Virginia, and Washington. These data include mostly the same information as the Monitoring of Federal Criminal Sentences data, but on the state court level as opposed to the federal. Unfortunately, different states included different combinations of defendant characteristic data (age, race, etc.) and some did not include this information at all.

The same prison capacity data was used for both datasets. The prison capacity data was aggregated from several sources. The first is the Census of State and Federal Adult Correctional Facilities, a survey taken every 5 years which includes the rated capacity of every adult correctional facility in the United States. Census years 1995, 2000, 2005, and 2012 were included. To fill in the years between census reports, data from individual state department of corrections websites was obtained. The amount of data available and its specific source varied by state. For example, Ohio data was collected from semi-annual audit reports of each facility in the state by the Correctional Institution Inspection Committee, whereas Arizona data was collected from capacity listings

in weekly prison population head counts performed by the Arizona Department of Corrections, Rehabilitation & Reentry⁸.

Some 60+ private prison contracts were received through a combination of state public records requests as well as a large batch acquired from In the Public Interest, “a nonprofit research and policy center that studies public goods”⁹. From these contracts it was determined which prisons had minimum occupancy clauses, and what percentage guarantee was in place. To calculate clause guaranteed capacity, each prison’s capacity was multiplied by the percentage level of beds that were guaranteed, and this number was added to the minimum occupancy clause capacity for that state. The remainder was included in the non-minimum occupancy clause capacity for that state.

Both the federal and state sentencing datasets were combined with the capacity data. In the federal dataset, states in which capacity data was available were included. These states are Alabama, Arkansas, Arizona, Colorado, Georgia, Kentucky, Maryland, Minnesota, Missouri, North Carolina, Nevada, Ohio, Oregon, Tennessee, Texas, Virginia, and Washington. In the state dataset, the states included were limited to those states in which both capacity data as well as state level sentencing data was available. These states are Arkansas, Georgia, Maryland, Texas, and Virginia. Both datasets were restricted to include only male defendants, and the capacity data was restricted to include only capacity for male inmates.

Both datasets have some form of limitation. The federal dataset is far more comprehensive and includes many more states and variables, however, the fact that it is federal sentencing interferes slightly with identification when using it for analysis. While federal criminals are certainly more likely to be sentenced to a prison in the same state where the crime and trial took place, federal courts are not under an explicit requirement to do so. There is a possibility that a prisoner sentenced in one state may be sent to another state to serve out their time. Therefore, the prison capacity of the state in which they were sentenced is not as directly related to their sentencing outcomes. The state level data handles this problem in that it only includes state court trials, however, the dataset loses its robustness through number of states included, treatment changes, and availability of controls. Only five states are included in total, and most critically only two of those states are home to private prisons that enact minimum occupancy clauses at some point in time. Even in those two states changes in that capacity happen only sporadically to a minute number of institutions. It is difficult to say with confidence that an effect of minimum occupancy clauses seen in only two states from a small handful

⁸ As a time-saver, pre-cleaned capacity data for Alabama, Arkansas, Colorado, Georgia, Indiana, Kentucky, Maryland, Minnesota, Missouri, North Carolina, Nevada, Oregon, Tennessee, Texas, Virginia, and Washington, was graciously provided by Christian Dippel and Michael Poyker. This data comes from the same sources listed previously.

⁹ <https://www.inthepublicinterest.org/about-us/>

of institutions is representative of its effect more generally. Furthermore, the defendant characteristic data is far less robust, preventing the inclusion of any of the controls that are included in the federal dataset. Defendant characteristics are an important component of any criminal justice analysis and being unable to include any such controls is a serious detriment to the robustness of the results from the state data. For these reasons, the results from the federal dataset should be weighted far more heavily than those from the comparatively weak state level dataset. That being said, results from the two datasets together, in addition with a few other specifications of states to be analyzed, have potential to shed light on the effects that minimum occupancy clauses have on sentencing, and thus the mechanism behind the effect that private prisons have on sentencing.

5. Re-establishing the Private Prison Effect

5.1 Empirical Model

The first step is to re-establish the effect of private prisons on sentencing in our dataset. While Dippel and Poyker (2020) and Galinato and Rohla (2020) identify this effect in more robust ways, it is still important to confirm that the same is seen in this data. It would make little sense to perform analysis on further specified effects under the assumption that the more general private prison effect exists when in the data when it in fact does not. I perform the following regression:

$$Y_{st} = B_0 + B_1 * Pub_{st} + B_2 * Priv_{st} + B_3X + State_s + Year_t + StateYear_{st} + e_{st}$$

Here, Pub_{st} is the public prison capacity in state s in year t , $Priv_{st}$ is the private prison capacity in state s in year t , X is a vector of characteristic controls (race, education, age). $State_s$ are state fixed effects, $Year_t$ are year fixed effects, and $StateYear_{st}$ are state-year pair fixed effects. It should be noted that for a given regression, either state fixed effects and year fixed effects are included, or state-year fixed effects are included. No one regression can include both sets of fixed effects. Particular attention will be paid to the models that include state-year fixed effects, as these capture variations between states in certain years, allowing different states to follow different trends over time. When the fixed effects are split into state and year separately, it must be assumed that there is one national time-based trend, and that each state has one specific effect associated with that state that cannot change from year to year¹⁰. e_{st} is the error term.

¹⁰ In models analyzing the extensive margin, state and year fixed effects must be separated. This is because each individual entry is in fact a state year pair, so state-year fixed effects are impossible.

Y_{st} takes different forms, depending on whether the intensive or extensive margin is being analyzed. For the intensive margin, Y_{st} is the total number of months a convicted person is sentenced to, as each observation in this regression is an individual in a state in a year who was dealt a certain sentence. For the extensive margin, Y_{st} takes the form $\ln(\text{Total_Persons_Sentenced})$, or the log of the total number of people sentenced in a given state in a given year¹¹. In this regression, an observation is a state in a year, and each is associated with the number of persons sentenced in that state in that year. In extensive margin regressions, individual characteristic controls X cannot be included as it is not individuals being analyzed.

5.2 Results

5.2.1 Federal Dataset

Intensive Margin

In table 1 are the results from four regressions. With state and year fixed effects plus an intercept, the private prison effect initially appears to be ever so slightly negative, though quite close to zero. With state-year fixed effects in model (2) the effect is insignificant. Our model of primary interest here is (4). This model includes the more robust state-year fixed effects, the same as model (2), though with the removal of the intercept term the fit of the model based on R^2 improves significantly. The coefficient on *Priv* of 0.066 means that each new private prison bed in a state will increase a sentenced person's term by 0.066 months, or about 2 days. This is a considerable effect considering the average new private prison has hundreds of new beds. All coefficients on public prison capacity are insignificant, as expected. On the intensive margin, the effect of private prisons on sentencing does appear in this dataset.

Extensive Margin

In table 2 are results from four more regressions. Here, our models of interest are (2) and (4), the level and log versions of the same regressions. The removal of the intercept term increases the model fit considerably compared their respective versions that include the intercept term. That being said, the removal of this intercept term does not actually change the coefficient on *Priv*. All models utilize state and year fixed effects. From a levels perspective, each new private prison bed is associated with an additional .003 persons sentenced per 100,000. The log model produces a coefficient of .0001, meaning a new private prison bed is associated with a .01% increase in the number of people sentenced per 100,000. As with the intensive margin, all

¹¹ A non-log specification of *Total_Persons_Sentenced* is also included, and both results are shown in the same table.

Table 1: Re-establishing private prison effect - Intensive (Federal)

	<i>Dependent variable:</i>			
	Sentence Total: Months			
	(1)	(2)	(3)	(4)
Priv	-0.001*** (0.0001)	-0.026 (0.019)	-0.001*** (0.0001)	0.066** (0.033)
Pub	0.0001 (0.0001)	-0.005 (0.003)	0.0001 (0.0001)	-0.005 (0.003)
State FE:	Yes	No	Yes	No
Year FE:	Yes	No	Yes	No
State-Year FE:	No	Yes	No	Yes
Intercept:	Yes	Yes	No	No
Race:	Yes	Yes	Yes	Yes
Age:	Yes	Yes	Yes	Yes
Education:	Yes	Yes	Yes	Yes
Observations	446,410	446,410	446,410	446,410
R ²	0.105	0.107	0.407	0.408
Adjusted R ²	0.105	0.107	0.407	0.408
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01	

Table 3: Re-establishing private prison effect - Intensive (State)

	<i>Dependent variable:</i>			
	Sentence Total: Months			
	(1)	(2)	(3)	(4)
Priv	-0.014*** (0.0004)	-0.094 (0.081)	-0.014*** (0.0004)	0.013 (0.045)
Pub	-0.002*** (0.0002)	-0.0001 (0.003)	-0.002*** (0.0002)	-0.0001 (0.003)
State FE:	Yes	No	Yes	No
Year FE:	Yes	No	Yes	No
State-Year FE:	No	Yes	No	Yes
Intercept:	Yes	Yes	No	No
Race:	No	No	No	No
Age:	No	No	No	No
Education:	No	No	No	No
Observations	1,425,630	1,425,630	1,425,630	1,425,630
R ²	0.054	0.059	0.092	0.096
Adjusted R ²	0.054	0.059	0.092	0.096
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01	

Table 2: Re-establishing private prison effect - Extensive (Federal)

	<i>Dependent variable:</i>			
	Persons Sentenced per 100,000			
	<u>Logs</u>		<u>Levels</u>	
	(1)	(2)	(3)	(4)
Priv	0.0001*** (0.00002)	0.0001*** (0.00002)	0.003*** (0.001)	0.003*** (0.001)
Pub	-0.00000 (0.00001)	-0.00000 (0.00001)	0.0005 (0.0004)	0.0005 (0.0004)
State FE:	Yes	Yes	Yes	Yes
Year FE:	Yes	Yes	Yes	Yes
Intercept:	Yes	No	Yes	No
Observations	272	272	272	272
R ²	0.744	0.988	0.746	0.881
Adjusted R ²	0.725	0.987	0.727	0.871
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01	

Table 4: Re-establishing private prison effect - Extensive (State)

	<i>Dependent variable:</i>			
	Persons Sentenced per 100,000			
	<u>Logs</u>		<u>Levels</u>	
	(1)	(2)	(3)	(4)
Priv	0.0001*** (0.00002)	0.0001*** (0.00002)	-0.023*** (0.006)	-0.023*** (0.006)
Pub	0.00004** (0.00002)	0.00004** (0.00002)	0.021*** (0.004)	0.021*** (0.004)
State FE:	Yes	Yes	Yes	Yes
Year FE:	Yes	Yes	Yes	Yes
Intercept:	Yes	No	Yes	No
Observations	64	64	64	64
R ²	0.992	0.999	0.995	0.997
Adjusted R ²	0.988	0.999	0.993	0.996
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01	

coefficients on public prison capacity are insignificant. While the extensive margin effect is not as pronounced as the one found on the intensive margin, the effect of private capacity on the number of people sentenced does exist in this dataset.

5.2.2 State dataset

Intensive Margin

The results from the state dataset are considerably murkier than those from the federal dataset, likely due to the drawbacks mentioned previously. In table 3, there are negative results for the first three regressions. Model (4) has the potential to be the most informative. It includes the more robust state-year fixed effects, and without the intercept term has the highest R^2 . The point estimate of the coefficient on *Priv* is positive, which is somewhat reassuring if the goal is to show that the private prison effect does exist. The standard error is certainly large enough that the true effect could swing negative, and though it has the highest R^2 of these four models, it is still incredibly low at .096. Models (1) and (3), which use state and year fixed effects, are the only ones with significant coefficients and both are slightly negative, though the fixed effects used in these models are not as robust. I certainly cannot claim that the positive effect on the intensive margin is necessarily present, though I must also stop short of saying there is certainly a negative effect either, due to the abysmally low R^2 scores of all models and the weakness of this dataset. From these regressions I cannot confirm that the intensive margin effect exists in the state dataset, which should be kept in mind when analyzing this data further.

Extensive Margin

In table 4 are regressions that examine the extensive margin. In both logs models (with and without intercept term) the coefficient on *Priv* is a positive .0001, is statistically significant, and the R^2 is incredibly high. This would imply that each new private prison bed is associated with a .01% increase in the number of persons sentenced per 100,000. The coefficient on *Pub* is more than two times smaller, at .00004, and is significant as well. Here, the effect of private prison beds is a great deal larger than the effect of public prison beds, which is exactly the effect expected. This result is called into question however by the results of the levels versions of these regressions. Both coefficients on *Priv* are a statistically significant -.023, which would imply that each private prison bed is associated with a *decrease* in the number of sentenced persons per 100,000 by .023. In these regressions the coefficient on *Pub* is a significant and positive .021. These results are the opposite of what was expected and what was found in the logs regressions. That the inclusion of the log component was enough to switch the sign on the coefficient of interest speaks to the small absolute size of these coefficients

and the fact that in this dataset the true effect is likely closer to zero. This is not completely surprising, as this dataset has far fewer states than the federal one as well as far fewer instances of treatment changes. As with what was seen on the intensive margin, all results from this dataset should be taken with a grain of salt and are likely not as robust as the more data intensive federal sentencing dataset.

6. Minimum Occupancy Clauses

Now that the effect of private prisons on sentencing in the data has been re-established (to a much higher degree in the federal dataset), I can begin analyzing the specific effect that minimum occupancy clauses have on sentencing, which is the true purpose of this paper.

Previously, the theoretical models predicted two potential effects of minimum occupancy clauses on sentencing. Under the cost saving model, the reduced (or should I say removed, even) marginal cost of sentencing to the judge would increase the sentence that the judge would consider optimal. Under the judicial capture model, the effect was a bit more ambiguous, but was likely to have the opposite effect, as in states where private prisons would successfully lobby for minimum occupancy guarantees they would have less incentive (and ability) to lobby for explicitly increased sentencing. The states where minimum occupancy clauses were instated were likely the states which had judicial systems that the private prisons identified as being difficult to “bribe” or “capture.” The minimum occupancy clauses were their best option in these states to maximize revenue but would likely not have the same effect on sentencing that could be produced by explicitly lobbying for increased sentencing in other states. Under this model, the effect of minimum occupancy clause capacity could be weaker than the effect of non-minimum occupancy clause capacity, even to the point of being zero or negative, while still being consistent with the model’s predictions. I now turn to the data to see which effect appears.

6.1 Empirical Model

Private prison capacity ($Priv_{st}$) is split into the capacity guaranteed by a minimum occupancy clause, and that which is not. A similar regression model as before is specified, the only change being the delineation of private prison capacity into the two categories:

$$Y_{st} = B_0 + B_1 * Pub_{st} + B_2 * PrivYes_{st} + B_3 * PrivNo_{st} + B_4X + State_s + Year_t + StateYear_{st} + e_{st}$$

$PrivYes_{st}$ is the private prison capacity that is guaranteed by a minimum occupancy clause in state s in year t . $PrivNo_{st}$ is the private prison capacity that is not guaranteed by a minimum occupancy clause in state s in year t . The rest of the terms are the same as before.

6.2 Results

6.2.1 Federal Dataset

Intensive Margin

Table 5 contains results from four regressions, with similar specifications as before. Our primary models of interest are (2) and (4), which utilize the more robust state-year fixed effects, especially model (4) which has the best fit due to the removal of our intercept term. Interestingly, in model (4) the coefficient on $PrivNo$ (the prisons without occupancy guarantees) is unchanged from the overall private prison coefficient in the aggregated model from before, at .066. The coefficient on $PrivYes$ has nearly doubled to .117, and both are statistically significant. Here, an additional private prison bed under a minimum occupancy guarantee will increase a defendant's sentence by .117 months, or nearly four whole days, which is nearly twice the effect of an additional private prison bed without a minimum occupancy guarantee. When the size of the average new private prison is considered, this effect is quite large.

Extensive Margin

In table 6 are the results from the regressions on the extensive margin. The choice to include the intercept or not does not change any coefficients. The coefficient on $PrivNo$ is a significant .0001, meaning that each new private prison bed without a minimum occupancy clause increases the number of persons sentenced per 100,000 by .01%, which again is the same as the coefficient on overall capacity seen previously. The coefficient on $PrivYes$ is however smaller, at .00003, though it is not significant. More informative here is the levels regression, which produces statistically significant coefficients on both terms. The coefficient on $PrivYes$ in both levels models is .004, and the coefficient on $PrivNo$ in both levels models is .003, meaning that each new private prison bed covered by a minimum occupancy clause is associated with .001 more persons being sentenced per 100,000 than a new private prison bed without a minimum occupancy clause. When considering the number of beds in many new prisons as well as how many 100,000s of people there are in certain states, this could have real implications on sentencing in places. As with the intensive margin, the effect is largely consistent with the theory from the cost saving model.

Table 5: Federal Data - Intensive

	<i>Dependent variable:</i>			
	Sentence Total: Months			
	(1)	(2)	(3)	(4)
PrivYes	-0.002*** (0.0001)	0.027** (0.012)	-0.002*** (0.0001)	0.117* (0.063)
PrivNo	-0.001*** (0.0001)	-0.026 (0.019)	-0.001*** (0.0001)	0.066** (0.033)
Pub	0.0001** (0.0001)	-0.005 (0.003)	0.0001** (0.0001)	-0.005 (0.003)
State FE:	Yes	No	Yes	No
Year FE:	Yes	No	Yes	No
State-Year FE:	No	Yes	No	Yes
Intercept:	Yes	Yes	No	No
Race:	Yes	Yes	Yes	Yes
Age:	Yes	Yes	Yes	Yes
Education:	Yes	Yes	Yes	Yes
Observations	446,410	446,410	446,410	446,410
R ²	0.105	0.107	0.407	0.408
Adjusted R ²	0.105	0.107	0.407	0.408

Note: *p<0.1; **p<0.05; ***p<0.01

Table 7: State Data - Intensive

	<i>Dependent variable:</i>			
	Sentence Total: Months			
	(1)	(2)	(3)	(4)
PrivYes	-0.013*** (0.0004)	-0.035 (0.039)	-0.013*** (0.0004)	0.014 (0.049)
PrivNo	-0.014*** (0.0004)	0.006 (0.009)	-0.014*** (0.0004)	0.008 (0.010)
Pub	-0.002*** (0.0002)	-0.0001 (0.003)	-0.002*** (0.0002)	-0.0001 (0.003)
State FE:	Yes	No	Yes	No
Year FE:	Yes	No	Yes	No
State-Year FE:	No	Yes	No	Yes
Intercept:	Yes	Yes	No	No
Race:	No	No	No	No
Age:	No	No	No	No
Education:	No	No	No	No
Observations	1,425,630	1,425,630	1,425,630	1,425,630
R ²	0.054	0.059	0.092	0.096
Adjusted R ²	0.054	0.059	0.092	0.096

Note: *p<0.1; **p<0.05; ***p<0.01

Table 6: Federal Data - Extensive

	<i>Dependent variable:</i>			
	Persons Sentenced per 100,000			
	<u>Logs</u>		<u>Levels</u>	
	(1)	(2)	(3)	(4)
PrivYes	0.00003 (0.00003)	0.00003 (0.00003)	0.004*** (0.001)	0.004*** (0.001)
PrivNo	0.0001*** (0.00002)	0.0001*** (0.00002)	0.003*** (0.001)	0.003*** (0.001)
Pub	-0.00000 (0.00001)	-0.00000 (0.00001)	0.0004 (0.0004)	0.0004 (0.0004)
State FE:	Yes	Yes	Yes	Yes
Year FE:	Yes	Yes	Yes	Yes
Intercept:	Yes	No	Yes	No
Observations	272	272	272	272
R ²	0.745	0.988	0.747	0.881
Adjusted R ²	0.724	0.987	0.727	0.871

Note: *p<0.1; **p<0.05; ***p<0.01

Table 8: State Data - Extensive

	<i>Dependent variable:</i>			
	Persons Sentenced per 100,000			
	<u>Logs</u>		<u>Levels</u>	
	(1)	(2)	(3)	(4)
PrivYes	0.0001** (0.00004)	0.0001** (0.00004)	-0.018* (0.011)	-0.018* (0.011)
PrivNo	0.0001*** (0.00002)	0.0001*** (0.00002)	-0.023*** (0.006)	-0.023*** (0.006)
Pub	0.00004** (0.00002)	0.00004** (0.00002)	0.022*** (0.004)	0.022*** (0.004)
State FE:	Yes	Yes	Yes	Yes
Year FE:	Yes	Yes	Yes	Yes
Intercept:	Yes	No	Yes	No
Observations	64	64	64	64
R ²	0.992	0.999	0.995	0.997
Adjusted R ²	0.988	0.998	0.993	0.996

Note: *p<0.1; **p<0.05; ***p<0.01

6.2.2 State Dataset

Intensive Margin

In table 7 the regressions of interest are (3) and (4), as removal of the intercept again improves model fit considerably. With state and year fixed effects all coefficients are significant, though their signs are somewhat unexpected¹². All capacity coefficients, *PrivYes*, *PrivNo*, and *Pub*, are negative. *PrivYes* is less negative than *PrivNo*, which in a roundabout way is somewhat consistent with the cost saving hypothesis¹³. While model (4) has no statistically significant coefficients, it is still interesting to analyze, as this model utilizes the more robust state-year fixed effects (and does have the highest R^2 , though they are all incredibly low). In this model, the point estimate on *PrivYes* is .014, meaning a new private prison bed would be associated with a sentence length increase of .014 months, or about a half of a day. This is nearly twice as high as the coefficient on *PrivNo*, of .008, which is higher still than the coefficient on *Pub*, a -.0001, which would effectively be zero even if it were statistically significant. While it of course cannot be overlooked that the standard errors on these coefficients are large enough that nothing can be said with confidence from these results, it is interesting to note that these coefficients are exactly what the cost saving model (which seems to be the leading candidate in the other results so far) would predict.

Extensive Margin

The results in table 8 experience the same odd behavior when adding logs as in the overall private prison capacity model. In a logs sense, the effect of *PrivYes* and *PrivNo* are the same at .0001, and both are higher than the *Pub* effect of .00004. However, when looking at levels, the coefficients on both *PrivYes* and *PrivNo* become negative. This strange behavior, as mentioned before, likely points towards the extensive margin effect in this dataset more realistically being zero since a mere log transformation was able to swap the sign. With fear of sounding repetitive, it should again be noted that the results from the state dataset should not be weighted as heavily due to severe data restrictions.

7. Minimum Occupancy Clause State Restriction

A further specification that may provide interesting results is to restrict the states that will be included in the sample to only those where a minimum occupancy clause was enacted at some point in time. In states where these clauses never appear, there may be other confounding factors that prevent their enactment and muddle

¹² This is less unexpected when you remember that this dataset's initial results were already questionable.

¹³ Though not really...

our results in some way. It is also possible, particularly in the state dataset, that the overwhelming number of observations where minimum occupancy clauses are not present dominate the data and obscure the true effect of the existence of minimum occupancy clauses.

In a theoretical sense, looking at only states that enact a minimum occupancy clause at some point is akin to the situation in the judicial capture model of states with a relatively low a (corruption of judges) and relatively high $L_i(c)$ (likelihood of successfully enacting minimum occupancy clause). Under this model with this specification of states, it would not be expected that minimum occupancy clauses would increase sentencing. If judges are unable to be bribed for increased sentencing, lobbying budget would likely flow outwards from these states once the minimum occupancy clause has been instituted. It would thus not be surprising, under the judicial capture hypothesis, to see the same effect between the two in these states, even to the point of being the same as that of public capacity increases. Under the cost saving hypothesis however, this would not be the case, and the increased effect of minimum occupancy clauses would likely appear more pronounced under this specification of sample states.

7.1 Results

7.1.1 Federal Dataset

Intensive Margin

In table 9, the primary model of interest is (4). As before, its fit is the most accurate due to the exclusion of the intercept, and it uses the more robust state-year fixed effects. The coefficient seen on *PrivYes* in this model is the largest yet, at a statistically significant .385. This implies that in these states, a single new private prison bed that is guaranteed by a minimum occupancy clause will increase a defendant's sentence by .385 months, or nearly 12 days. This is a colossal effect given the size of most new private prisons. Interestingly, in this model the effect of non-occupancy clause private prison beds is negative and quite large (in absolute terms) as well. The negative value on the *PrivNo* coefficient as well as possibly the incredibly large size of the *PrivYes* coefficient may to some extent be quirks of the restricted nature of this dataset specification. It is possible that the effect of private prisons with occupancy clauses are so strong in these specific states that the effect of the non-occupancy clauses becomes relatively negative as a byproduct of the empirical model (since all states in this dataset now have some form of minimum occupancy clause).

Table 9: Federal Data - Intensive
Only states with occupancy quota at some point in time

	<i>Dependent variable:</i>			
	Sentence Total: Months			
	(1)	(2)	(3)	(4)
PrivYes	-0.0005** (0.0002)	-0.092 (0.122)	-0.0005** (0.0002)	0.385*** (0.055)
PrivNo	0.00003 (0.0002)	-0.228* (0.126)	0.00003 (0.0002)	-0.228* (0.126)
Pub	-0.0005*** (0.0001)	-0.016*** (0.003)	-0.0005*** (0.0001)	-0.016*** (0.003)
State FE:	Yes	No	Yes	No
Year FE:	Yes	No	Yes	No
State-Year FE:	No	Yes	No	Yes
Intercept:	Yes	Yes	No	No
Race:	Yes	Yes	Yes	Yes
Age:	Yes	Yes	Yes	Yes
Education:	Yes	Yes	Yes	Yes
Observations	132,146	132,146	132,146	132,146
R ²	0.169	0.170	0.458	0.459
Adjusted R ²	0.168	0.169	0.458	0.458

Note: *p<0.1; **p<0.05; ***p<0.01

Table 11: State Data - Intensive
Only states with occupancy quota at some point in time

	<i>Dependent variable:</i>			
	Sentence Total: Months			
	(1)	(2)	(3)	(4)
PrivYes	0.0004 (0.0005)	-0.060*** (0.011)	0.0004 (0.0005)	0.014 (0.011)
PrivNo	-0.0003 (0.001)	-0.005*** (0.0003)	-0.0003 (0.001)	0.006** (0.003)
Pub	0.004*** (0.001)	-0.0001 (0.001)	0.004*** (0.001)	-0.0001 (0.001)
State FE:	Yes	No	Yes	No
Year FE:	Yes	No	Yes	No
State-Year FE:	No	Yes	No	Yes
Intercept:	Yes	Yes	No	No
Race:	No	No	No	No
Age:	No	No	No	No
Education:	No	No	No	No
Observations	788,201	788,201	788,201	788,201
R ²	0.091	0.091	0.412	0.412
Adjusted R ²	0.091	0.091	0.412	0.412

Note: *p<0.1; **p<0.05; ***p<0.01

Table 10: Federal Data - Extensive
Only states with occupancy quotas at some point in time

	<i>Dependent variable:</i>			
	Persons Sentenced per 100,000			
	<u>Logs</u>		<u>Levels</u>	
	(1)	(2)	(3)	(4)
PrivYes	0.00005 (0.00004)	0.00005 (0.00004)	0.005** (0.002)	0.005** (0.002)
PrivNo	0.0001* (0.00004)	0.0001* (0.00004)	0.005** (0.002)	0.005** (0.002)
Pub	0.00001 (0.00002)	0.00001 (0.00002)	0.0004 (0.001)	0.0004 (0.001)
State FE:	Yes	Yes	Yes	Yes
Year FE:	Yes	Yes	Yes	Yes
Intercept:	Yes	No	Yes	No
Observations	80	80	80	80
R ²	0.848	0.989	0.752	0.859
Adjusted R ²	0.831	0.988	0.724	0.841

Note: *p<0.1; **p<0.05; ***p<0.01

Table 12: State Data - Extensive
Only states with occupancy quotas at some point in time

	<i>Dependent variable:</i>			
	Persons Sentenced per 100,000			
	<u>Logs</u>		<u>Levels</u>	
	(1)	(2)	(3)	(4)
PrivYes	0.00002 (0.0001)	0.00002 (0.0001)	0.010 (0.030)	0.010 (0.030)
PrivNo	0.00003 (0.0001)	0.00003 (0.0001)	0.010 (0.032)	0.010 (0.032)
Pub	0.00003 (0.0001)	0.00003 (0.0001)	0.018 (0.034)	0.018 (0.034)
State FE:	Yes	Yes	Yes	Yes
Year FE:	Yes	Yes	Yes	Yes
Intercept:	Yes	No	Yes	No
Observations	16	16	16	16
R ²	1.000	1.000	0.999	1.000
Adjusted R ²	0.997	1.000	0.996	0.999

Note: *p<0.1; **p<0.05; ***p<0.01

Extensive Margin

In table 10 are results from the extensive margin, again restricted to only states that enacted minimum occupancy clauses at some point. In levels terms (regressions (3) and (4)) the effects are the same, with each new private prison bed, clause or not, being associated with .005 additional persons sentenced per 100,000. This is one of the only results from the federal dataset that appears in line with the judicial capture theory as opposed to the cost saving theory. In logs, the effect of *PrivYes* is insignificant and effectively zero, while the effect of *PrivNo* is a significant .0001. That *PrivNo* has a stronger effect on the number of people sentenced than *PrivYes*, (and especially that the effect of *PrivYes* is approximately equal to zero) *would be* in line with the judicial capture theory that minimum occupancy clauses are enacted in states where lobbying for additional sentencing is relatively difficult, and thus in states where they are not enacted are states where the private prisons can lobby directly for increased sentencing, and thus a stronger effect is seen on private prison beds without the minimum occupancy clause than those with it. Negating this interpretation however is that the stronger effect of non-minimum occupancy clause private prison capacity is seen when the data is restricted to *only* states that at some point enact a minimum occupancy clause. In the theory, this stronger effect of non-clause prison capacity would have to be coming from states with relatively low $L_i(c)$ from the model, and thus places where these clauses were not enacted¹⁴.

7.1.2 State Dataset

Intensive Margin

In table 11 I perform the same restriction, but this time on the state dataset. The primary model of interest is (4), as it is the model with state-year fixed effects and the best fit. The results are in line with what the cost saving theory would predict. The highest coefficient is *PrivYes*, with a point estimate of .014. It is not statistically significant, but the standard error is not incredibly egregious. The coefficient on *PrivNo* is a significant .006, just under half of the effect on *PrivYes*. The coefficient on *Pub* brings up the rear at an insignificant -.0001. Private prison beds under minimum occupancy clauses in these states are associated with an increase of a defendant's sentence of about half a day, while a private prison bed with no contract will only be associated with an increase in a defendant's sentence by about one fifth of a day.

¹⁴ All of this being said, it should again be noted that this restriction is somewhat arbitrary and done mostly out of curiosity. The most telling results still come from analysis using all states available.

Extensive Margin

The results from table 12 are not all too meaningful. In the logs regression, the effect of *PrivYes* is just barely smaller than that of *PrivNo* and *Pub* (.00002 vs .00003 and .00003). All are miniscule in an absolute value sense and none of them are significant. In the levels regression, the *PrivYes* and *PrivNo* coefficients are the same in both models at .010, and again nothing is significant. These results are not completely unexpected from an already weak dataset that was restricted down to merely two states.

8. Conclusion

8.1 Discussion

As mentioned previously, the most important results to focus on are those from the federal dataset, most specifically the intensive margin models with state-year fixed effects and the best fit, as well as the extensive margin models. Apart from a few extensive margin cases where both effects were largely identical or were indistinguishable from zero, the effect on sentencing of private prisons with minimum occupancy clauses was consistently stronger than the effect of private prisons without minimum occupancy clauses, which was in turn stronger than the effect of public prisons (which was consistently equal to zero). This is exactly what the theoretical model of the cost saving hypothesis would predict. That the majority of the most robust results fall directly in line with this theory is strong evidence that the cost saving hypothesis is the driving mechanism through which private prisons have incentivized harsher sentencing outcomes.

While the use of private prisons has recently been discontinued, this has happened before and been reversed, and so their use could be re-instated once again. It is not a guarantee that private prisons have seen their last days in the United States. Furthermore, private detention facilities are still being utilized by ICE and the U.S. Customs and Border Protection for the detainment of those classified as illegal immigrants. The incentive structure analyzed in this paper is thus still quite relevant today. Understanding the incentive mechanisms at play and how to properly guard against their abuse is crucial for ensuring a fair and just judicial system.

8.2 Future Work

There are of course areas where the analysis in this paper could be improved in future research. Over the next few months, I will continue to receive state level data from many states Departments of Corrections that have already been requested through different public records request processes. Once this data has been acquired, which includes county codes for the court system where sentencing took place, it will be possible to perform the more robust contiguous border county-pairs utilized by Dippel and Poyker (2020) to initially identify the

private prison effect on sentencing, and which was originally developed by Dube, Lester, and Reich (2010). My personal goal is to continually update the analysis in this paper as more and better data is received. With this added data the state dataset will reach an equal or greater level of robustness as the federal dataset has currently and will also then remove the slight identification shortfall present in the federal dataset. Better data and the more accurate identification procedure of using contiguous border county-pairs will allow the effect of minimum occupancy clauses to be specified quite accurately.

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